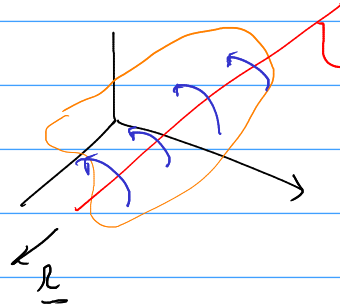
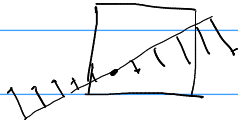
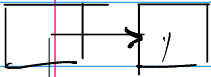


$$\epsilon_{ijk} = \epsilon_{kij}$$

$$\underline{\Omega} = \text{curl } \underline{v} = \nabla \times \underline{v}$$



Eje de rotación

$$\underline{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -3x_2 \\ 3x_1 - x_3 - 5 \\ x_2 \end{bmatrix} \Rightarrow x_2 = 0$$

$$3x_1 - x_3 - 5 = 0$$

$$\begin{cases} 3x_1 - x_3 = 5 \end{cases}$$

$$\begin{cases} 3x_1 - x_3 = 5 \\ x_2 = 0 \end{cases}$$

$$V_{ij} = \frac{1}{2} \left( \frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) \Rightarrow \begin{cases} V_{11} = 0 \\ V_{22} = 0 \\ V_{33} = 0 \\ V_{12} = 0 \\ V_{13} = 0 \\ V_{23} = 0 \end{cases}$$

$$\Omega_i = \left( \epsilon_{ijk} \frac{\partial v_k}{\partial x_j} \right)_i$$

$$\Omega_1 = \epsilon_{123} \frac{\partial v_3}{\partial x_2} + \epsilon_{132} \frac{\partial v_2}{\partial x_3} =$$

$$= 1 \cdot 1 - 1 \cdot (-1) = 2$$

$$\Omega_2 = \epsilon_{213} \frac{\partial v_3}{\partial x_1} + \epsilon_{231} \frac{\partial v_1}{\partial x_3} = 0$$

$$\Omega_3 = \epsilon_{312} \frac{\partial v_2}{\partial x_1} + \epsilon_{321} \frac{\partial v_1}{\partial x_2} =$$

$$= 1 \cdot 3 - (-3) = 6$$

Forma alternativa de determinar el vector de rotación

$$\underline{\Omega} = \begin{bmatrix} 2 \\ 0 \\ 6 \end{bmatrix}$$

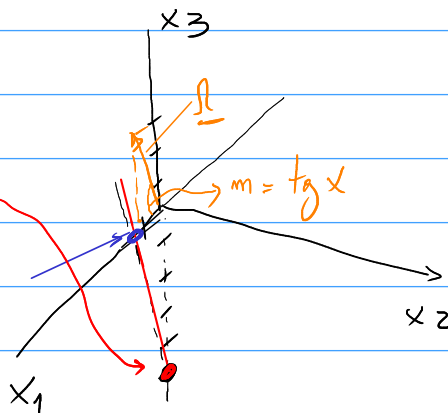
$$\underline{\Omega} = 2 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$$

$$x_3 = 3x_1 - 5$$

$$3x_1 - x_3 = 5$$

$$x_1 = 0 \Rightarrow x_3 = -5$$

$$x_3 = 0 \Rightarrow x_1 = 5/3$$



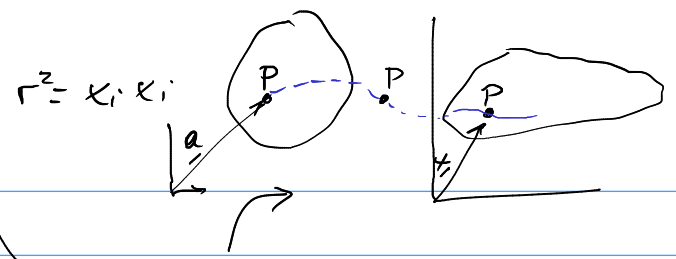
$$T(a_1, a_2, a_3, t)$$

$$T(a, t) = \frac{a^2 t^3}{\sqrt{a_2^2 + 4a_3}}$$

$a$  no depende del tiempo

$$\dot{T} = \frac{a^2}{\sqrt{a_2^2 + 4a_3}} \cdot 3t^2$$

$$T(\underline{x}, t) = \frac{3A t^4}{r} + t$$



$$\left. \frac{\partial T}{\partial \underline{x}} \right|_{\underline{x} = \underline{c}}$$

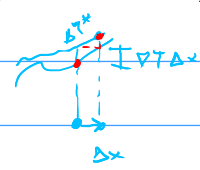
derivada temporal espacial (fija la posición)

$$\dot{T} = \frac{dT}{dt} = \frac{DT}{Dt}$$

derivada temporal material (sigue la partícula)

$$\dot{T} = \frac{dT(x_i(t), t(t))}{dt} = \left. \frac{\partial T}{\partial x_i} \right|_{t=fijo} \cdot \frac{dx_i}{dt} + \left. \frac{\partial T}{\partial t} \right|_{\underline{x}=fijo} \cdot \frac{dt}{dt}$$

$$T(\underline{x}, t)$$



$$\dot{T} dt \approx \Delta T = \nabla T \cdot \underline{\Delta x} + \Delta T|_{\underline{x}=fijo}$$

Ec de equilibrio diferencial:  $\rightarrow$  fuerza m6nica  $\frac{F_{22}}{\text{unidad de masa}} \cdot \rho \left[ \frac{U \cdot \text{densidad}}{\text{volumen}} \right] = \frac{F_{22}}{\text{volumen}} = X_i$

$\rightarrow \rho \cdot b_i$

$$\frac{\partial \pi_{ij}}{\partial x_j} + X_i = 0$$

$$\frac{\partial \pi_{ij}}{\partial x_j} u_i + X_i \cdot u_i = 0$$

$$\int_V \frac{\partial \pi_{ij}}{\partial x_j} u_i + \int_V X_i u_i = 0$$

$$\int_V \frac{\partial (A_{jke} \dots)}{\partial x_i} dV = \int_{\Gamma} A_{jke} \dots n_i d\Gamma$$

$$\int_V \frac{\partial (A_{jke} \dots)}{\partial x_j} dV = \int_{\Gamma} A_{jke} \dots n_j d\Gamma$$

$$\frac{\partial (\sigma_{ij} u_i)}{\partial x_j} = \frac{\partial \sigma_{ij}}{\partial x_j} u_i + \sigma_{ij} \frac{\partial u_i}{\partial x_j} \Rightarrow \frac{\partial \pi_{ij}}{\partial x_j} u_i = \frac{\partial (\pi_{ij} u_i)}{\partial x_j} - \pi_{ij} \frac{\partial u_i}{\partial x_j}$$

$$\int_V \frac{\partial (\pi_{ij} u_i)}{\partial x_j} - \int_V \pi_{ij} \frac{\partial u_i}{\partial x_j} + \int_V X_i u_i = 0$$

invariancia de  $\pi$   
 $T_i = \pi_{ij} n_j$   
 $= \pi_{kj} n_j$

$$\int_{\Gamma} \pi_{ij} u_i n_j - \dots = 0$$

$$\int_{\Gamma} T_i u_i - \int_V \pi_{ij} \frac{\partial u_i}{\partial x_j} + \int_V X_i u_i = 0$$

$$\pi_{ij} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

$$\pi_{ij} \frac{1}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right)$$

$$\pi_{ij} \frac{1}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \pi_{ij} \frac{1}{2} \left( \frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right)$$

$\epsilon_{ij}$

$$\frac{1}{2} \left( \pi_{ij} \frac{\partial u_i}{\partial x_j} - \pi_{ij} \frac{\partial u_j}{\partial x_i} \right)$$

$\rightarrow$  invariancia de  $\pi$

$$\frac{1}{2} \left( \pi_{ij} \frac{\partial u_i}{\partial x_j} - \pi_{ji} \frac{\partial u_j}{\partial x_i} \right)$$

$$\int_V \pi_{ij} \cdot \epsilon_{ij} = \int_{\Gamma} T_i u_i + \int_V X_i u_i$$